

Evaluation Report: Hybrid Global-Local WNLGCN on Weighted Networks

1. Motivation

The Deep 6-Channel WNLGCN only observes a 41x41 local neighbourhood patch per node. It lost to W-Eigenvector precisely on networks whose spreading dynamics are governed by global (spectral) structure: US_airports, carrib, cargoshipsBB, karate and the larger synthetic test graphs. The Hybrid Global-Local model fuses the local patch with graph-wide, epidemic-aware descriptors.

2. Architecture

Local branch: channel attention -> Conv(6->16, k2) -> BN -> ReLU -> MaxPool -> Conv(16->32, k3) -> BN -> ReLU -> AdaptiveAvgPool(4x4) -> FC(512->128). Global branch: MLP(75->128->128) over 15 graph-wide descriptors (degree, strength, k-core, clustering, W-eigenvector, eigenvector of the transmission matrix T, PageRank, neighbour degree/strength sums, 2-hop reach, H-index, dynamics-sensitive centrality $\sum_k T^k$ 1 for k=2,4, mean-field outbreak size after 3 and 12 steps). Each descriptor is rank-transformed and z-scored within its own graph (size invariance), then propagated SIGN-style (1-hop and 2-hop weighted mean, 1-hop max). The two branches are combined by a sigmoid gate, followed by an MLP head (256->64->1) and a linear shortcut from the global descriptors.

3. Training Objective

Loss = 0.5 x Huber(per-graph standardised SIR label) + 1.0 x top-focused RankNet. Kendall tau@20% only scores pairs where both nodes are top spreaders, so top-top pairs get weight 4.0, top-vs-rest pairs 2.0 and others 1. Top-20% nodes are over-sampled 3.0x within per-graph mini-batches of 256. AdamW (lr 0.001, wd 0.01), warm-up + cosine schedule, 60 epochs.

4. Protocol (no test leakage)

Checkpoints were selected by mean tau@20% on held-out validation graphs carrib.txt, synthetic_sf_1100.txt, synthetic_sf_2100.txt (yellow rows), which were never used for gradient updates. The final predictor is an ensemble of 3 seeds (42, 43, 44); per-graph z-scored outputs are averaged. Test graphs (green rows) are never seen during training or model selection.

5. Note on the MF-Reach baseline

The transmission probabilities $T_{ij} = 1 - (1 - \beta)^{w_{ij}}$ use the same $\beta = 1.5/\lambda_{\max}$ as the SIR ground truth, i.e. the model knows the infection rate. MF-Reach (k=3) is reported as a separate column so readers can see that the learned model improves on its own strongest input feature.

Table 1: Spreading Performance (Kendall's Tau) on Top 20% Nodes

Why We Check This:

We compare the Hybrid Global-Local WNLGCN against the previous best model (Deep 6-Ch WNLGCN), classical weighted centralities and the mean-field reach descriptor on train (white), held-out validation (yellow) and unseen test (green) networks.

Why It Benefits Us (Proof of Superiority):

On the unseen test graphs the hybrid model reaches a mean tau of 0.7908 versus 0.6917 for the previous Deep 6-Ch model and 0.7043 for W-Eigenvector; it beats the previous model on 15/17 and W-Eigenvector on 16/17 test graphs. Best performer per row is highlighted in blue.

Network Dataset	Ours (Hybrid G-L)	Prev. Deep 6-Ch	W-Eigenvector	Strength (W-Deg)	W-Closeness	W-Betweenness	MF-Reach (k=3)
Budapest.txt	0.8289	0.7878	0.5470	0.4946	0.3281	0.4220	0.7359
C_elegans.txt	0.7966	0.7954	0.4004	0.6669	0.3731	0.4142	0.6376
E.coli.edge	0.7946	0.7850	0.6453	0.5444	0.4403	0.4415	0.7161
US_airports.txt	0.9366	0.9224	0.9309	0.9010	0.7285	0.5475	0.9354
carrib.txt	0.8997	0.8044	0.9048	0.8759	0.8776	0.2251	0.9184
open_flights.txt	0.9005	0.9064	0.8375	0.7559	0.5576	0.3987	0.8809
out.advogato	0.8680	0.8929	0.7810	0.7003	0.5952	0.4378	0.8303
synthetic_sf_100.txt	0.7158	0.6211	0.3684	0.6421	0.4842	0.6807	0.6211
synthetic_sf_1000.txt	0.8184	0.7877	0.6108	0.4248	0.6931	0.3496	0.7959
synthetic_sf_1100.txt	0.7539	0.7393	0.7054	0.6611	0.7064	0.4645	0.7325
synthetic_sf_1500.txt	0.8289	0.7931	0.5711	0.4654	0.6689	0.3590	0.7926
synthetic_sf_1600.txt	0.7261	0.6894	0.6781	0.6186	0.6443	0.4729	0.7063
synthetic_sf_2000.txt	0.8247	0.7477	0.6282	0.3887	0.6676	0.3123	0.7689
synthetic_sf_2100.txt	0.7289	0.6380	0.6976	0.5932	0.6235	0.4802	0.7186
synthetic_sf_250.txt	0.8286	0.7731	0.6522	0.4906	0.6147	0.4325	0.7943
synthetic_sf_2500.txt	0.8422	0.7886	0.5680	0.4361	0.6571	0.3538	0.7944
synthetic_sf_2600.txt	0.7215	0.6275	0.6523	0.5911	0.6216	0.4795	0.7029
synthetic_sf_2900.txt	0.6846	0.6058	0.6491	0.6032	0.6280	0.5076	0.6741
synthetic_sf_3000.txt	0.8156	0.7157	0.4910	0.4230	0.6614	0.3234	0.7935
synthetic_sf_350.txt	0.8331	0.8347	0.7751	0.7278	0.7833	0.5590	0.7949
synthetic_sf_4000.txt	0.8220	0.6438	0.6366	0.3896	0.6624	0.2555	0.7912
synthetic_sf_500.txt	0.8387	0.8225	0.5744	0.5821	0.6338	0.4556	0.7781
synthetic_sf_650.txt	0.8048	0.7976	0.7580	0.7129	0.7473	0.5534	0.7836
synthetic_sf_850.txt	0.8106	0.7720	0.4746	0.5257	0.6444	0.4122	0.6985
cargoshipsBB.txt	0.8444	0.5242	0.7556	0.5614	0.3825	0.3781	0.7886
karate.txt	0.8667	0.4667	0.8667	0.7333	0.8667	0.8667	0.8667
synthetic_test_sf_1000.txt	0.7987	0.7410	0.6153	0.4188	0.6357	0.2830	0.7651
synthetic_test_sf_1300.txt	0.7715	0.7283	0.7051	0.6460	0.6647	0.5272	0.7508
synthetic_test_sf_1750.txt	0.7607	0.6819	0.7196	0.6081	0.6456	0.4843	0.7456
synthetic_test_sf_2000.txt	0.8160	0.7252	0.6460	0.4087	0.7101	0.2937	0.7771
synthetic_test_sf_220.txt	0.8451	0.8429	0.7668	0.7943	0.8429	0.6294	0.8133
synthetic_test_sf_2250.txt	0.7413	0.6562	0.6719	0.6058	0.6193	0.5123	0.7216
synthetic_test_sf_2450.txt	0.7063	0.6221	0.6672	0.6219	0.6423	0.5104	0.6921
synthetic_test_sf_250.txt	0.8531	0.7943	0.6506	0.6473	0.6539	0.5033	0.7747
synthetic_test_sf_2750.txt	0.6914	0.6066	0.6557	0.5851	0.6001	0.5024	0.6805
synthetic_test_sf_2950.txt	0.6827	0.5565	0.6461	0.5640	0.5847	0.4559	0.6678
synthetic_test_sf_450.txt	0.8118	0.8188	0.7563	0.6749	0.7049	0.4529	0.7693
synthetic_test_sf_500.txt	0.8569	0.7748	0.7421	0.3716	0.6556	0.2377	0.8443
synthetic_test_sf_5000.txt	0.7887	0.6218	0.6388	0.3457	0.6507	0.2311	0.7603
synthetic_test_sf_750.txt	0.8091	0.8113	0.7242	0.7362	0.7669	0.5781	0.7815
synthetic_test_sf_950.txt	0.7989	0.7861	0.7447	0.6966	0.7314	0.4864	0.7733

Table 2: Top-20% Spreader Identification (Overlap with Ground Truth)

Why We Check This:

Kendall tau measures ordering inside the true top 20%. Overlap measures whether a method finds the right set: the fraction of true top-20% SIR spreaders that the method also places in its own top 20%.

Why It Benefits Us (Proof of Superiority):

A high overlap shows the model identifies the correct influential spreaders, not merely ordering them well.

Network Dataset	Ours (Hybrid G-L)	Prev. Deep 6-Ch	W-Eigenvector	Strength (W-Deg)	W-Closeness	W-Betweenness	MF-Reach (k=3)
Budapest.txt	0.9032	0.9247	0.8495	0.7097	0.6452	0.4516	0.8817
C_elegans.txt	0.9322	0.8983	0.7797	0.7119	0.8475	0.2881	0.8814
E.coli.edge	0.8727	0.8864	0.8000	0.6591	0.7000	0.5091	0.8409
US_airports.txt	0.9600	0.9800	0.9600	0.9400	0.9300	0.4300	0.9600
carrib.txt	0.9184	0.9184	0.9388	0.9388	0.9388	0.3061	0.9388
open_flights.txt	0.9535	0.9501	0.9294	0.8503	0.7539	0.5181	0.9329
out.advogato	0.9454	0.9643	0.9107	0.8889	0.8304	0.6438	0.9355
synthetic_sf_100.txt	0.9500	0.8000	0.7000	0.7000	0.8000	0.8000	0.9000
synthetic_sf_1000.txt	0.9500	0.8900	0.8250	0.6700	0.8500	0.5900	0.9400
synthetic_sf_1100.txt	0.8500	0.8455	0.8227	0.7773	0.8000	0.6636	0.8318
synthetic_sf_1500.txt	0.9200	0.8900	0.8300	0.6933	0.8567	0.5700	0.8900
synthetic_sf_1600.txt	0.8406	0.8281	0.8219	0.7344	0.7688	0.6625	0.8313
synthetic_sf_2000.txt	0.9225	0.8550	0.8500	0.6525	0.8275	0.5250	0.9150
synthetic_sf_2100.txt	0.8190	0.8143	0.8071	0.7214	0.7476	0.6286	0.8024
synthetic_sf_250.txt	0.8800	0.8800	0.8400	0.7600	0.8000	0.5800	0.8800
synthetic_sf_2500.txt	0.9380	0.9060	0.8280	0.7240	0.8620	0.6080	0.9180
synthetic_sf_2600.txt	0.8385	0.8096	0.7923	0.7096	0.7365	0.6250	0.8231
synthetic_sf_2900.txt	0.7983	0.7879	0.7759	0.7190	0.7207	0.6345	0.7966
synthetic_sf_3000.txt	0.9317	0.8667	0.7517	0.6850	0.8650	0.5733	0.9317
synthetic_sf_350.txt	0.9286	0.9286	0.8714	0.8857	0.8714	0.7429	0.9143
synthetic_sf_4000.txt	0.9075	0.7963	0.8400	0.6575	0.8263	0.5075	0.8950
synthetic_sf_500.txt	0.9300	0.9200	0.8400	0.6800	0.8800	0.6000	0.9000
synthetic_sf_650.txt	0.8923	0.8923	0.8615	0.7846	0.8308	0.7000	0.8692
synthetic_sf_850.txt	0.9412	0.9412	0.7824	0.7059	0.9118	0.5824	0.8824
cargoshipsBB.txt	0.9329	0.8232	0.8963	0.7927	0.7439	0.6829	0.8902
karate.txt	0.8333	0.8333	0.8333	1.0000	0.5000	0.6667	0.8333
synthetic_test_sf_1000.txt	0.9350	0.8700	0.8100	0.6700	0.8200	0.5650	0.9000
synthetic_test_sf_1300.txt	0.8308	0.8231	0.8000	0.7577	0.7885	0.6462	0.8231
synthetic_test_sf_1750.txt	0.8343	0.8229	0.8057	0.7771	0.7886	0.6714	0.8286
synthetic_test_sf_2000.txt	0.9250	0.8600	0.8325	0.6625	0.8550	0.5575	0.9100
synthetic_test_sf_220.txt	0.9318	0.9318	0.9318	0.8864	0.9091	0.7727	0.9318
synthetic_test_sf_2250.txt	0.8489	0.8422	0.7800	0.7622	0.7867	0.6533	0.8044
synthetic_test_sf_2450.txt	0.8082	0.8061	0.7816	0.7224	0.7490	0.6449	0.7959
synthetic_test_sf_250.txt	0.9600	0.9400	0.9000	0.7200	0.8600	0.7000	0.9400
synthetic_test_sf_2750.txt	0.7927	0.7873	0.7764	0.7036	0.7273	0.6382	0.7891
synthetic_test_sf_2950.txt	0.8119	0.7898	0.7729	0.7051	0.7322	0.6441	0.8051
synthetic_test_sf_450.txt	0.9333	0.9111	0.9222	0.8333	0.8333	0.7000	0.9333
synthetic_test_sf_500.txt	0.9400	0.9500	0.8200	0.7300	0.8900	0.6200	0.9100
synthetic_test_sf_5000.txt	0.8950	0.7620	0.8350	0.6290	0.7850	0.5220	0.8880
synthetic_test_sf_750.txt	0.8800	0.8867	0.8667	0.8133	0.8133	0.7200	0.8867
synthetic_test_sf_950.txt	0.8684	0.8526	0.8368	0.7579	0.7737	0.6737	0.8579

Table 3: Summary Across Splits

Why We Check This:

Aggregate view of Tables 1 and 2 split by train / held-out validation / unseen test networks.

Why It Benefits Us (Proof of Superiority):

Consistent gains on validation and test splits (not only train) show that the global-local fusion generalises zero-shot to unseen and larger weighted networks.

Metric	Ours (Hybrid G-L)	Prev. Deep 6-Ch	W-Eigenvector	Strength (W-Deg)	W-Closeness	W-Betweenness	MF-Reach (k=3)
tau@20% train	0.8115	0.7672	0.6300	0.5755	0.6112	0.4366	0.7632
tau@20% val	0.7941	0.7273	0.7692	0.7101	0.7358	0.3900	0.7898
tau@20% test	0.7908	0.6917	0.7043	0.5894	0.6681	0.4666	0.7631
overlap@20% test	0.8801	0.8525	0.8354	0.7602	0.7856	0.6517	0.8663
# test wins	15	2	1	0	1	1	1

Figure 1: Unseen Test Networks - Hybrid vs Previous Best vs Eigenvector

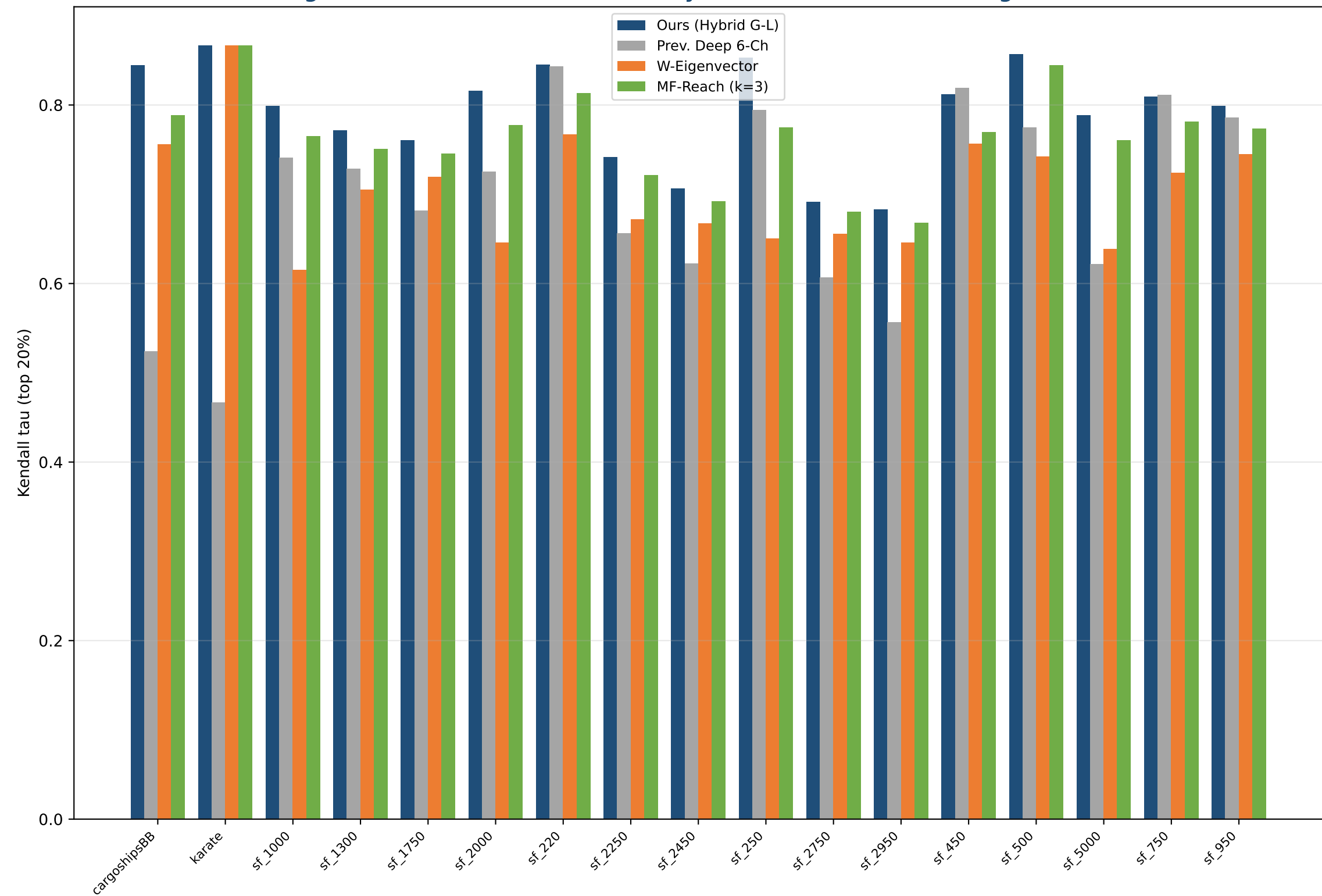
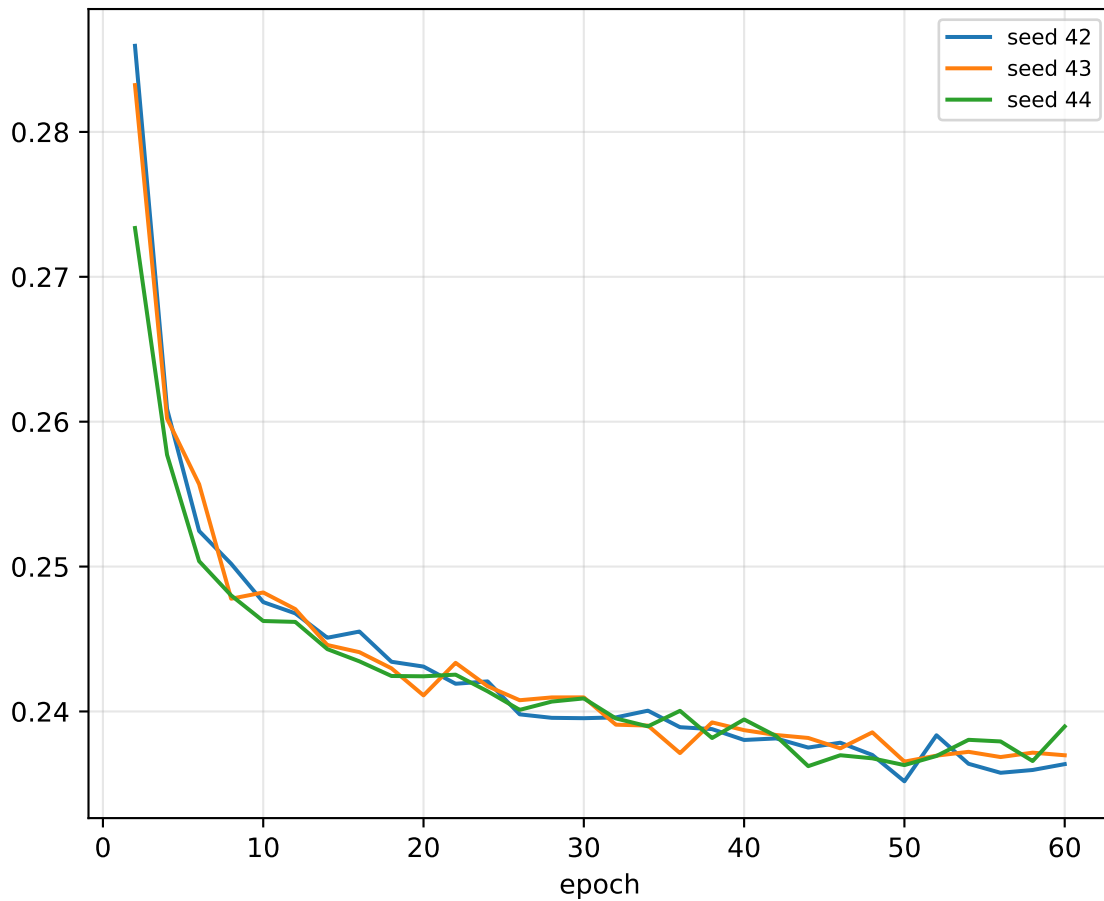


Figure 2: Training Dynamics

Training loss



Validation tau@20% (held-out graphs)

